INCH-POUND

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SUPERSEDING
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28 August 1981

MILITARY SPECIFICATION

LOADER, SCOOP TYPE, DED, 4 x 4

ARTICULATED FRAME STEER,

2-1/2 CUBIC YARD, COMMERCIAL CONSTRUCTION EQUIPMENT

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a commercial model, articulated frame steer, four-wheel-drive, diesel-engine driven, 8,000-pound operating load capacity loader with a 2-1/2 cubic yard, multipurpose, front-end mounted, scoop type bucket.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1108	- Extinguisher, Fire, Vaporizing Liquid.
VV-F-800	- Fuel Oil, Diesel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-1283	- Can, Gasoline, Military, 5-Gallon.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-T-3351	- Tractor, Full-Tracked, Low Speed; Tractor Wheeled, Agricultural; and Tractor Wheeled, Industrial; and their Attachments, Packaging of.
MIL-M-7866	- Molybdenum Disulfide, Technical, Lubrication Grade.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-I-14511	- Insulation Sheet, Cellular, Plastic: Thermal.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw, NATO Code Number G-354, Metric.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-R-46164	- Rustproofing for Military Vehicles and Trailers.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-B-46176	- Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative.
MIL-C-62218	- Corrosion Preventive Compounds, Cold-Application (For New and Fielded Motor Vehicles and Trailers).
MIL-T-62314	- Test Equipment (Simplified) for Internal Combustion Engines - Reprogramable (STE/ICE-R) Test Set.
MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits For Army Materiel.
MIL-STD-1791	- Designing for Internal Aerial Delivery in Fixed Wing Aircraft.
MS35424	- Light, Marker, Clearance-Blackout, 24 Volt.
MS51318	- Headlight: Blackout, 24 Volt, Waterproof.
MS51330	- Stop Light - Taillight, Vehicular, 24 Volt, Blackout Tail, Blackout Stop.

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MS52131 - Connector, Plug, Electrical Intervehicle Power Cable.
MS53052 - Bracket Assembly, Liquid Container, Five Gallon.

HANDBOOKS

MILITARY

MIL-HDBK-113 - Guide for the Selection of Lubricants, Functional Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DEPARTMENT OF THE ARMY

MTMCTEA Pamphlet 70-1 - Transportability for Better Strategic Mobility.

(Application for copies should be addressed to the Military Traffic Management Command, Transportation Engineering Agency (MTMC/TEA), ATTN: MTT-TR, P.O. Box 6276, Newport News, VA 23606-0276.)

DEPARTMENT OF LABOR

OSHA 29 CFR 1926 - Occupation Safety and Health Administration.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

DRAWINGS

TACOM

12258941 - Connector, Receptacle.
12258955 - STE/ICE Vehicle Test Card.

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the US Army Tank Automotive Command, ATTN: AMSTA-MD, Warren, MI 48397-5000.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Commodities on Open-Top Cars.

(Application for copies should be addressed to the Association of American Railroads, American Railroads Building, 50 F Street, NW, Washington, D.C. 20001.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J53	- Minimum Performance criteria for Emergency Steering of wheeled Earthmoving Construction Machines.
SAE J209	- Instrument Face Design and Location for Construction and Industrial Equipment.
SAE J231	- Minimum Performance Criteria for Falling Object Protective Structure (FOPS).
SAE J276	- Steering Frame Lock for Articulated Loaders and Tractors.
SAE J386	- Operator Restraint Systems for Off-Road Work Machines.
SAE J514	- Hydraulic Tube Fittings.
SAE J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
SAE J678	- Speedometers and Tachometers-Automotive.
SAE J732	- Specification Definitions-Loaders.
SAE J742	- Capacity Rating-Loader Bucket.
SAE J818	- Operating Load for Loaders.
SAE J833	- USA Human Physical Dimensions.
SAE J898	- Control Locations for Off-Road Work Machines.
SAE J899	- Operator's Seat Dimensions for Off-Road Self-Propelled Work Machines.
SAE J925	- Minimum Service Access Dimensions for Off-Road Machines.
SAE J994	- Alarm-Backup-Electric-Performance, Test, and Application.
SAE J1040	- Performance Criteria for Rollover Protective Structures (ROPS) for Construction, Earthmoving, Forestry, and Mining Machines.
SAE J1048	- Symbols for Motor Vehicle Controls, Indicators and Tell-Tales.
SAE J1152	- Braking Performance-Rubber-Tired Construction Machines.
SAE J1349	- Engine Power Test Code-Spark Ignition and Diesel.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order or precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The loader shall be the latest model of the standard product of the contractor. It shall have been marketed and in commercial field use in significant numbers, equipped with a multipurpose bucket in the 2.50 cubic yard size for at least 1 year prior to the invitation for bids or request for proposal. Product improvements incorporated in the commercial production units during this 1-year period are acceptable. When specified requirements are not stated herein, all items listed as standard equipment in the contractor's published specification brochures and catalogs and delivered to commercial customers as standard equipment shall be furnished. The loader shall be equipped with optional equipment as specified herein. Optional equipment is defined as equipment not standard with the loader but which has been furnished to the commercial customer, such as special features or attachments. The loader and attachments shall withstand usage encountered in military operations such as loading trucks and stockpiling material, as well as excavating undisturbed and compacted soil without breakage, malfunction or permanent deformation. The loader shall conform to all Federal laws and regulations governing safety, and pollution, which are in effect for the type of equipment at the time of manufacture. For definition of terms used herein, SAE J732 shall apply. The loader shall be rubber-tired, diesel-engine driven, four-wheel-drive with rear axle oscillation and shall have an articulated frame steering system. The loader shall be equipped with a hydraulically operated, front mounted, scoop type, multipurpose bucket (see 3.19.1) and shall be used as a clamshell to handle irregularly shaped objects and for general bulldozer work. It shall also be used as a scraper.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Production loader. All loaders furnished under this specification shall be physically and mechanically identical to the first article loader that is approved by the Government, except that product improvements incorporated in the contractor's standard product lines are permitted at no cost to the Government subject to the following:

a. Physical and performance characteristics have not been degraded.
b. New components involving Government-selected repair parts are directly interchangeable with the superseded component or part.
c. The Government is notified of all changes to any item on the recommended parts list where fit and function is effected by a minimum of 45 days prior to the production date of the change. When the changes and alterations have been incorporated in the production loader, the Government shall be furnished the production effectivity by serial number and dates.
d. Appropriate changes are made to the Government validating operating, maintenance, and parts manuals before delivery.

3.4 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

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3.4.1 Material deterioration prevention and control. The loader shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the loader may be exposed.

3.4.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the loader may be newly fabricated from recovered materials to the maximum extent practicable, provided the loader produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the loader.

3.5 Cab. A fully enclosed, removable operator's cab (or ROPS cab) with door(s) windows, and front and rear windshield shall be furnished. The doors of the cab shall have a latch mechanism that will latch the door(s) in the fully open position. Front windshield wiper(s) and windshield washer system shall be furnished. The cab shall be insulated against cold and shall be sound suppressed conforming to 3.27. A heater shall be furnished and shall have a rating of not less than 18,000 Btu per hour. Ventilation fans or defrosters shall be furnished that direct airflow toward all glazing. Each fan shall have an "on-off" switch and be directionally adjustable. Quick-disconnect fittings shall be provided as applicable to facilitate cab removal. The cab shall have slinging eye(s) to aid in removing the cab. Outside mirrors mounted on left and right side of the cab shall be furnished. The cab shall be provided with drain holes where applicable to eliminate the accumulation of water. A fire extinguisher bracket shall be provided and mounted in the operator compartment. The bracket shall be Coast Guard approved and shall secure a 5 pound Halon fire extinguisher as described by A-A-1108.

3.5.1 Controls. All controls for the loader shall be located in the operator's compartment and shall be within the reach of the operator in his normal operating position conforming to requirement 3.26. To indicate the function or use, all controls shall be identified by plates either on or adjacent to each control.

3.6 Rollover protective structure (ROPS) and falling objects protective structure (FOPS). The loader shall be equipped with a bolted down or pin-connected ROPS and FOPS structure which may be integral with the fully enclosed cab (see 3.5). The structure shall be removable and replaceable without cutting or welding. The ROPS and FOPS shall conform to SAE J1040 and SAE J231, respectively. The ROPS shall also comply with applicable OSHA 29 CFR 1926. A

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seatbelt conforming to SAE J386 shall be properly installed for use by the operator. Quick-disconnect fittings shall be provided as applicable to facilitate ROPS and FOPS structure (ROPS cab) removal. Slinging eye(s) conforming to MIL-STD-209 shall be provided as attaching points for ROPS and FOPS structure (ROPS cab) removal.

3.7 Fenders. Front and rear fenders shall be furnished. Rear fenders may be an integral part of the rear housing or structure.

3.8 Seat. The contractor's standard commercial operator's seat shall be provided. The seat shall be located to provide the operator when seated with seatbelt fastened and properly adjusted convenient access to operating controls and visibility of the work zone and all instruments and gages mounted on the instrument panel.

3.9 Horn. A conventional operator-controlled horn shall be furnished.

3.10 Backup alarm. A backup alarm shall be provided on the loader and shall comply with SAE J994, type C. The backup alarm signal shall be 20 dB(A) above the normal loader operating noise.

3.11 Engine. The diesel engine shall meet the requirements of MIL-STD-1410 for application in a class II end item. The diesel engine shall be the engine furnished commercially, including all systems, components, accessories, and auxiliaries. The engine shall be equipped with a muffler. The exhaust stack shall be equipped with a rain cap or shall be fabricated and mounted to prevent the entry of rain or snow. The engine shall be the same engine that has been used in the manufacturer's standard commercial loader of the same size, for at least 1 year prior to contract award, and shall be of USA manufacture. The diesel engine shall operate on fuel conforming to both VV-F-800, grade DF 2 and MIL-T-83133, grade JP-8.

3.11.1 Engine horsepower. The applied net horsepower of the engine at the flywheel, when measured shall comply with SAE J1349, and shall be not less than 130 horsepower.

3.11.2 Engine side panels. The loader engine shall be equipped with removable side panels with quick-disconnect latches, horizontally or vertically hinged engine compartment doors, or a hinged "tilt away" type engine enclosure.

3.11.3 Air cleaner. A dry-type air cleaner with a precleaner and automatic dust ejector shall be furnished.

3.11.4 Engine cooling system. The contractor's standard or optional cooling system meeting the requirements of 3.28 shall be provided.

3.11.5 Tachometer drive. A heavy-duty key-type tachometer drive shall be provided and shall have a 0.1562-inch female connection conforming to SAE J678. The output drive shall be covered with a threaded captive chained cap for protection while not in use.

3.11.6 Oil sampling valve. An oil sampling valve shall be provided on the engine and transmission. The valve shall be manually operated and shall automatically close when released. It shall be made of material resistant to

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corrosion so it will not contaminate the sample. The discharge part of the valve shall be covered with a captive chained cap conforming to SAE J514. The valve shall be located to ensure that the personnel will not be exposed to danger when taking oil samples with the engine running. The oil tap shall be located to provide samples that are a true representation of oil that is flowing while the engine is running. Sampling valves shall be labeled crankcase or transmission, whichever is appropriate, to indicate the type of oil sampled.

3.12 Drive train.

3.12.1 Transmission. A full power shift transmission shall be furnished. The transmission shall provide not less than three forward speeds and three reverse speeds. When a dual range, twin turbine torque converter is furnished, not less than two forward gears and one reverse gear shall be provided, resulting in four forward speeds and two reverse speeds. The loader shall be equipped with a transmission that automatically disconnects the drive train when the service brakes are applied. The loader shall also have the capability of having the service brake applied without disconnecting the drive train. The transmission must be placed in neutral before the engine can be started and the loader set in motion.

3.12.2 Torque converter. The loader shall be equipped with a torque converter. A torque converter oil cooler shall be furnished.

3.12.3 Drive axles. The front and rear drive shall have differentials and planetary final drives. A limited slip differential or a hydraulic differential lock shall be furnished in the front or rear axle. Torque transferring differentials, of the torque bias type, shall be furnished in the front and rear axles or front axle only.

3.13 Fuel tank. The fuel tank shall be a type normally furnished as standard equipment with a capacity for 10 hours of continuous normal operation. The tank shall be equipped with a means for draining water and sediment from the bottom of the tank. Fuel capacity shall be verified during conduct of the construction tasks (see 4.5.2.2). The fuel fill port shall be labeled "DIESEL FUEL ONLY" or "JP-8 FUEL ONLY", whichever is appropriate.

3.14 Steering and braking.

3.14.1 Steering system. The loader shall have an articulated frame type steering system with a minimum of 30 degrees of articulation in each direction. The steering system shall provide full control of the loader under all normal operating conditions and shall provide emergency steering, meeting the requirements of SAE J53.

3.14.2 Braking system. A foot controlled service braking system shall be provided with disc brakes acting on all four wheels. A parking system and an emergency stopping system shall be provided. The braking system shall comply with SAE J1152. When nonpetroleum-based hydraulic brake fluid is used in a hydraulic braking system, it shall be of the silicone type conforming to MIL-B-46176. Asbestos material shall not be used for brake shoes or lining on this vehicle.

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3.14.2.1 Air brake system. When an air brake system is provided; an alcohol evaporator with shatter resistant translucent container, or air dryer system with automatic drain valve, shall be provided.

3.14.3 Hydraulic system. The loader shall be equipped with the contractor's standard hydraulic system which shall provide sufficient force for operating all attachments and related components. An adjustable automatic boom kickout and an automatic bucket positioner shall be furnished.

3.15 Tires. Tires shall be of the traction type tread (L-2), nylon, tubeless, with not less than a 12-ply rating. Tires shall be of the wide base type, size 20.5 by 25. All tires shall be of the same size, type, and ply rating. Tires furnished to the Government shall not be more than 12 months old from date of cure of the rubber to date of acceptance by the Government.

3.16 Electrical system. The loader shall be equipped with the contractor's standard or optional electrical system which meets the requirements specified herein. A 24-volt starting system shall be furnished.

3.16.1 Lighting. The following minimum lights with associated circuits, circuit breakers, or other protective devices and switches shall be provided:

a. Four front work and travel lights.

b. Two combination stop and taillights.

c. Two rear work lights.

d. Turn signals with emergency flashers.

e. Instrument panel lighting or backlighted gauges.

The brake lights shall overide the emergency flashers.

3.16.2 Alternator. The contractor's high output option alternator shall be furnished. The rated output shall be not less than 50 amps.

3.16.3 Instruments and indicators. The following instruments and indicators shall be provided.

a. Engine oil pressure gage, warning light, or diode monitoring system.

b. Engine coolant temperature gage, warning light, or diode monitoring system.

c. Torque converter oil temperature gage, warning light, or diode monitoring system.

d. Ammeter or voltmeter, warning light, or diode monitoring system.

e. Engine hourmeter or service meter.

f. Air pressure gage (when equipped with air system).

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g. Low-air-pressure warning buzzer (when equipped with air system).

h. Fuel level indicator (dipstick or gage).

i. Hydraulic oil level indicator (dipstick, sight tube, or gage).

j. Air cleaner restriction indicator.

Additional instruments and indicators standard with the contractor's commercial practice, which contribute to the safety and reliability of loader operation, shall be furnished.

3.17 Battery. The battery or batteries shall be the contractor's standard or optional batteries for cold weather starting as specified in 3.28.

3.17.1 Battery enclosure. A battery enclosure shall be provided for the battery or batteries. The enclosure shall be treated with an acid-resistant coating on the interior of the enclosure. Drain holes shall be provided to eliminate the accumulation of acids. There shall be a warning placed on the enclosure to warn personnel of the presence of toxic fumes and the explosive hazards.

3.17.2 Slaving components. The loader shall be equipped with a 24-volt slave receptacle conforming to MS52131. The slave receptacle shall permit charging of the loader batteries and slave starting of the engine from an external power source. It shall also provide a power source for charging and slaving other equipment. The slave receptacle shall be installed on the exterior of the loader near the battery enclosure and shall be accessible to personnel standing on the ground. The voltage potential of the slave receptacle shall be identified on a plate adjacent to the receptacle. The plate shall conform to MIL-P-514 or contractor's standard plate, subject to the approval of the contracting officer.

3.17.3 Fungus- and moisture-resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I with 1 percent copper 8-quinolinolate (by weight) or 7 percent salicylanilide based on the nonvolatile content of varnish.

a. Components or circuit elements that are inherently fungus-and-moisture resistant or which are hermetically sealed need not be treated.

b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both, to give a minimum dry-film thickness or 1 mil to component or element surfaces previously cleaned and prepared so the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.18 Towing device. The loader shall be equipped with a fixed-type drawbar, including a drawbar pin and a positive pin retaining means.

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3.19 Attachments.

3.19.1 Multipurpose bucket. The loader shall be furnished with a multipurpose-type bucket. The SAE nominal heaped bucket capacity rating shall be 2.50 cubic yards when rated conforming to SAE J742. The bucket shall be equipped with not less than eight bolt-on teeth. The bucket width shall be at least 1 inch greater than the loader without bucket. The bucket shall operate as a front loader, clamshell, dozer, and scraper.

3.19.2 Counterweight. Removable rear counterweight may be furnished on the loader to improve stability and to meet the requirements of this specification. The counterweight furnished shall be the same as that furnished by the contractor on his standard commercial loader when equipped with a 2.50 cubic yard multipurpose bucket. Tire ballast is not acceptable and shall not be furnished.

3.20 Special tools. Unless otherwise specified (see 6.2), any nonstandard or special tools required for equipment operation and operator maintenance of the loader shall be furnished with the loader.

3.20.1 Toolbox. A lockable storage compartment(s) or toolbox of sufficient size to hold all tools and equipment required for operation and operator maintenance shall be provided. The toolbox shall be provided with holes in the bottom for drainage.

3.21 Special requirements.

3.21.1 Blackout lighting. A blackout lighting system shall be furnished to meet the following requirements:

a. Each vehicle shall be equipped with the following blackout lights:

(1) One left-front-corner marker light conforming to MS51318.

(2) Two front-corner marker lights conforming to MS35424-1.

(3) Two rear-corner marker and stop taillights conforming to MS51330.

b. The blackout system will be controlled by an interior switch, readily accessible to the driver, which will prevent accidental disengagement of the blackout system from the blackout mode.

c. All regular exterior lights and horn, if electric, will be made inoperable automatically when in the blackout mode.

d. The interior lighting will have a variable-brightness control switch readily accessible to the driver.

e. A separate wiring harness shall be provided for the blackout lights only if the required blackout lights cannot be powered using the standard wiring harness.

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f. The blackout lights shall be oriented in the appropriate direction for roading of the loader. The lights shall be positioned at the following locations:

(1) Blackout driving lamp: mounted between the center and left-hand side of the vehicle, as far forward and as near the line of the operator's vision as practicable at a height of approximately 48 inches above the road surface.

(2) Front blackout marker lamps: one on each side of the vertical centerline, as far apart as practicable and at the same level between 15 and 72 inches above the road surface.

(3) Rear corner marker and stop taillights: one on each side of the vertical centerline, as far apart as practicable and at the same level between 15 and 72 inches above the road surface.

Changes to the locations outlined above will be subject to the approval of the contracting officer.

3.21.2 Winterized loader. When specified (see 6.2), the loader shall be winterized to permit starting and operation in arctic climates at ambient temperatures to -50 ^{0}F and storage at ambient temperatures to -60 ^{0}F. The contractor shall furnish one loader for examination and testing within the timeframe specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce winterized loaders that comply with the requirements of this specification.

3.21.3 Winterization kit. The kit shall consist of the components for preheating loader engine coolant and battery box(s); also insulated sound suppressed ROPS cab, hot water personnel heater (equipped with one or more shut-off valve(s) in the coolant lines to stop and vary the flow of coolant to the heater), defrosting system for cab windows, front and rear windshield wipers, radiator winterfront, insulated engine and radiator compartments, insulated battery box(s), battery finned connectors, and an alcohol evaporator when an air brake system is furnished. Winterization components shall not affect the performance of the loader and all other loader requirements shall be met as specified herein.

3.21.4 Kit installation. The winterized loader shall be capable of performing loading operations.

3.21.5 Insulation. Insulation shall comply with MIL-I-14511, 0.50 inch. Adhesive for bonding insulation shall retain its adhesive properties at ambient temperatures to -60 ^{0}F.

3.21.6 Heater and heat exchanger. The Eberspacher "Swingfire" heater, Part No. 26-9002, or equal, shall be utilized for preheat purposes. When specified (see 6.2), a quantity of heaters as specified (see 6.2) shall be furnished. The Eberspacher heat exchanger (water jacket), Part No. 26-9013, or equal that is compatible with the Swingfire heater, shall be provided for heating the engine coolant. The heat exchanger shall be located near the engine compartment for easy accessibility for insertion of heater. The heater unit starting operation shall be provided for insertion of heater. The heater unit starting operation

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shall be provided by the loader's 24-volt electrical system, using a wiring harness with Bosh Receptacle, Part No. 135050012, or equal, compatible with the heater starter-cable connector. After heater start and insertion into heat exchanger, preheating of the engine coolant shall be provided by thermosiphon action. The hot air from the heater exhaust shall be routed to the battery box(s) for overall hot-air drenching of the battery box(s) and then routed away from battery box(s) to the atmosphere. The hot air flow shall not be impeded by excessive back pressure or leakage. To retain the hot air, the system shall be insulated and heat shielded, as required.

3.21.7 Kit performance. The winterization kit shall be designed so the following criteria can be met when the loader has been cold soaked for 24 hours at a temperature of -50 ^{0}F:

a. Engine start within 5 minutes after preheating engine coolant and battery box(s) for 90 minutes.

b. Within 1 hour after personnel heater start, the heater shall have raised and stabilized the cab temperature to a desired +40 ^{0}F in the area occupied by the operator's torso, when seated.

c. Within 1 hour after personnel heater start, the defrosting system shall have cleared sufficient glassed area for safe loader operation, as well as visibility of the loader bucket and work area normally seen when there is no frost on the windows.

d. All loader controls shall be operational.

e. The transmission and brake controls shall be functional.

3.21.8 Diagnostic connector assembly (DCA). Unless otherwise specified (see 6.2), the loader shall incorporate an easily accessible DCA in the operator's cab for interface with the Government's Simplified Test Equipment/Internal Combustion Engine (STE/ICE) specified in MIL-T-62314, appendix B. All requirements for the DCA shall conform to TACOM Drawing 12258941. As a minimum, the DCA shall be capable of monitoring the functions defined in table II of Report CR82-588-003 (see 6.4), and the following: alternator/generator output voltage, alternator/generator field voltage, alternator/generator negative cable voltage drop, fuel solenoid voltage, starter voltage, starter current first peak and average starter circuit resistance, battery internal resistance, battery internal resistance change, fuel supply pressure and fuel filter restriction (Delta P). Additional test parameters to be monitored in either the DCA or Transducer Kit (TK) mode shall be as follows: fuel return pressure, fuel filter pressure drop, engine coolant temperature, starter solenoid voltage, battery resistance change, and transmission oil pressure. Test points identified to monitor functions in the TK mode shall be accessible without the removal of major assemblies and installed attachments. Those additional test parameters which are inaccessible for monitoring in the TK mode shall be monitored in the DCA mode. Fittings and adapters required for interface with the STE/ICE test set when monitoring in the TK mode, shall be permanently installed on the loader to assure that adapters in the STE/ICE test set readily mate with TK mode test point locations. A separate wiring harness shall be provided for the DCA and shall include all wiring and necessary hardware to perform required capabilities. A fuel shutoff method shall be provided for conducting compression imbalance

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checks. When technical manuals are furnished, the contractor shall also provide STE/ICE vehicle test cards in the format identified in TACOM Drawing 12258955,addressing both DCA and TK measurements and shall incorporate the test cards into the loader technical manuals, Data Item Description DI-TMSS-80527 applies for this requirement (see 6.2.2).

3.22 Antivandalism protection. The loader shall be equipped with locks to secure the following:

a. Fuel tank filler cap.

b. Hydraulic oil reservoir filler cap.

c. Transmission oil filler cap.

d. Engine oil filler cap.

e. Radiator filler cap.

f. Toolbox.

g. Battery box.

h. Cab doors.

When any of the above items are secured by lockable doors or lockable engine side panels, individual locks are not required. Keys for ignition, cab doors, cover panels, filler caps, and tool and battery boxes shall be interchangeable to the maximum extent possible. Two sets of keys shall be furnished with each loader and each set shall be identical for every loader.

3.23 Lubricants. The procedure for the selection of lubricants shall conform to MIL-STD-838, section 5. Lubricants selected shall conform to MIL-HDBK-113, chapter 2. When the specification of the lubricant selected includes a requirement for a qualified products list (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.23.1 Lubrication. A lubrication means shall be provided for all moving parts requiring lubrication. Each loader shall be serviced and run-in with military lubricants conforming to MIL-L-2104, MIL-L-2105, MIL-L-46167, MIL-M-7866, MIL-G-10924 and MIL-G-23827 as applicable. Initial servicing and run-in may be accomplished with lubricants standard to the contractor's commercial practice (except for the braking system), if compatible with military lubricants.

3.24 Performance. The loader shall have the following performance capabilities:

a. Lift, lower, and dump-rated loads.

b. Load the bucket to the SAE-rated heaped capacity (SAE J742).

c. Load stockpiled material into dump trucks.

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d. Dig and transfer stockpiled materials.

e. Excavate undisturbed and compacted soil.

f. As a clamshell, handle irregularly shaped objects.

g. As a dozer, perform backfilling and general bulldozer work. Dozer segment shall not interfere with the digging position of the bucket when bank loading when the bucket cutting edge is horizontal.

h. As a scraper, load the bucket to its SAE-struck capacity in undisturbed clay-loam soil (SAE J742).

3.24.1 Operating load. The loader shall have an SAE-rated operating load of 8000 pounds (SAE J818).

3.24.2 Tipping load. The loader shall have a full-turn tipping load SAE rating of not less than 16,000 pounds (SAE J732).

3.24.3 Lift capacity. The loader shall have an SAE lift capacity at the carry position of not less than 20,000 pounds. The loader shall have an SAE lift capacity to maximum height of not less than 10,000 pounds (SAE J732).

3.24.4 Breakout force. The loader shall have an SAE breakout force of not less than 17,500 pounds (SAE J732).

3.24.5 Clam closure force. The closure force at the lip of the closed clam shall be not less than 7000 pounds.

3.24.6 Drawbar pull. The empty loader shall develop a drawbar pull of not less than 16,000 pounds at 1.5 mph at the specified operating weight, on a dry, level, concrete surface, with not more than a 15-percent slip factor.

3.24.7 Travel speed. The empty loader shall be capable of a forward travel speed of not less than 19 mph on level pavement.

3.24.8 Cycle time. The loader shall have an SAE raising time of not more than 7.5 seconds, an SAE lowering time of not more than 5.5 seconds, and an SAE dump time of not more than 4.0 seconds (SAE J732).

3.24.9 Clearance circle. The loader shall have an SAE clearance circle diameter of not more than 46 feet (SAE J732).

3.24.10 Stability. The loader shall negotiate 27-percent side slopes and 50-percent longitudinal slopes without any wheel of the loader losing contact with the ground. All parts requiring lubrication shall be lubricated when the loader is operating under these conditions.

3.24.11 Fording. The loader shall start and operate during and after fording operations in fresh water in any depth up to 20 inches, including wave action without damage, and without leakage of water into reservoirs containing lubricants.

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3.25 Human factors. The characteristics of the loader with ROPS cab shall allow for operation and maintenance by the 5th to 95th percentile soldier conforming to MIL-STD-1472, SAE J833 and SAE J925, to the extent commercially available. Operating controls shall be located within the maximum control space conforming to MIL-STD-1472 and SAE J898 to the extent commercially available. The operator's seat dimensions shall conform to SAE J899. Instrumentation shall comply with SAE J209. Controls shall be provided with universal symbols conforming to SAE J1048.

3.26 Safety. All available standard or optional accessories shall be included to provide maintenance and operating personnel protection from rotating or reciprocating parts and from all parts subject to high operational temperatures. Exhaust discharge shall be directed to not endanger personnel. All platforms and steps shall have antiskid surfaces. Contractor's standard grabirons shall be provided. A decal shall be furnished and shall be located in the area of the articulation point, warning of limited personnel clearance during frame articulation and placement of steering frame lock for the loader operating and transport modes. No asbestos material shall be used in the manufacture of this vehicle.

3.27 Noise limits. The noise produced by the loader shall conform to MIL-STD-1474 requirements, with the exception of MIL-STD-1474, 5.2, 5.3, and 5.4 when tested in accordance with 4.5.2.20. The provision of MIL-STD-1474, 4.3 and 4.4, shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the contracting officer. Hazard signs shall conform to MIL-STD-1474, 4.3, except the sign will state "HEARING PROTECTION REQUIRED WITHIN (specify) FEET." The sign shall be readable at the distance indicated on the sign.

3.28 Environmental requirements. Without use of any external means, equipped with only standard or optional equipment, the engine shall start within 5 minutes. The loader shall operate under full-load conditions within 15 minutes after engine start in any ambient temperatures from -25 to +120 ^{0}F. Cold start aids must be integral with the loader and no aids may be used which rely upon external support such as an electrical power source. Handheld aerosol cans are not permitted. The loader shall be capable of operating in hot, humid, tropical environments and dry, dusty environments. The loader shall be capable of being stored for extended periods in temperatures ranging from -30 to +160 ^{0}F without damage.

3.29 Decontamination unit. The decontamination unit shall have a bracket assembly conforming to MS53052 and shall be mounted by the contractor. The location of the bracket shall be such that it is accessible to personnel and provides clearance to remove and emplace a 5-gallon container conforming to MIL-C-1283.

3.30 Radio interference suppression. The loader shall be suppressed to limit electromagnetic radiation to limits of SAE J551.

3.31 Transportability. The loader shall be capable of being transported by military or commercial trailers, trains, marine vessels, aircraft and must be capable of withstanding the impact forces encountered in shipment without damage or permanent deformation. The loader shall be equipped with tiedown or slinging

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provisions. A steering frame lock conforming to SAE J276 shall be provided. Complete diagrams and instructions for lifting the loader and tying down the loader on freight cars and truck trailers shall be furnished in the operator's manual. The operator's manual shall also include instructions for component removal when required for transport.

3.31.1 Tiedown provisions. The tiedown provisions shall conform to MIL-STD-209, class 2 or 3, type II equipment, and to MIL-STD-1791 for equipment restraint criteria. The tiedown provisions shall satisfactorily complete the pull testing as specified without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components.

3.31.2 Slinging provisions. The slinging provisions shall conform to MIL-STD-209, class 1 or 3, type II. The provisions shall enable the complete loader to be lifted in the normal operating position. The provisions shall be located so that not less than one inch clearance is maintained between slings and all exterior parts and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. Slinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.31.1. All slinging/tiedown provisions shall be labeled "LIFT", "TIEDOWN", or "LIFT TIEDOWN", as appropriate, in 1-inch (2.54 cm) high letters.

3.31.3 Air transportability. The loader shall meet the requirements of MIL-STD-1791 for air transport. When required (see 6.2), the contractor shall provide sufficient data to the Government in the Technical Information Package to confirm that the loader being supplied can be loaded, transported in the C-130, C-141, and C-5A aircrafts.

3.31.4 Rail transportability. The loader shall be rail transportable in CONUS and NATO countries without restrictions. The loader shall have a dimensional profile within the Gabarit International de Chargement (GIC) in accordance with MTMC/TEA pamphlet 70-1, outline diagram, when loaded on a 50-inch (127 cm) high rail car. The loader shall be capable of withstanding shock loads resulting from rail impact testing conforming to 4.5.2.24.3 without failure, damage, or permanent deformation.

3.31.5 Highway transportability. The loader, when loaded on a semitrailer/tractor, shall be within the highway permit limits of all States.

3.32 Treatment and painting. The portions of the loader normally painted, internally and externally, shall be cleaned, treated, and painted conforming to MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), top coat color shall be camouflage green 383 conforming to MIL-C-46168. When specified (see 6.2), the loader shall be overcoated conforming to the Government-furnished camouflage patterns with MIL-C-46168 top coat of the colors specified in the camouflage patterns, except the first article may be camouflage green 383. The vehicles shall be undercoated with Corrosion Preventive Compounds in accordance with MIL-C-62218, these compounds shall be applied in accordance with MIL-R-46164.

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3.32.1 Camouflage pattern data. When specified (see 6.2), the contractor shall provide camouflage pattern data.

3.33 Physical characteristics.

3.33.1 Dimensions. The loader shall conform to the dimensions shown in table I (SAE J732 - a through g).

TABLE I. Dimensions.

Characteristic	Requirement
a. Height to hinge pin at full lift, minimum.	140 inches
b. Overall height, maximum.	138 inches
c. Overall length, maximum.	312 inches
d. Forward dumping dump height measured to the bucket cutting edge at full lift and 45-degree dump angle, minimum.	100 inches
e. Forward dumping reach measured to bucket cutting edge at full lift and 45-degree dump angle, minimum.	31 inches
f. Maximum rollback at carry position, minimum.	40 degrees
g. Digging depth, minimum.	1.5 inches
h. Ground clearance, minimum.	14 inches
i. Maximum clamshell opening, minimum.	46 inches
j. Bucket as a dozer, depth of cut with pitch angle of dozer cutting edge 50 degrees from the horizontal, minimum.	2 inches
k. Rear axle oscillation (unless articulation joint allows for oscillation), each direction, minimum.	10.5 degrees

3.33.2 Weight. The fully serviced and fueled loader, equipped as specified herein with bucket, ROPS and FOPS and cab, but with any add-on type rear counterweight removed, shall weigh not less than 25,000 pounds. The fully serviced and fueled loader equipped with bucket, ROPS and FOPS and cab and with all rear counterweights installed, shall weigh not more than 33,500 pounds.

3.34 Identification plates. Each loader shall be furnished with an identification plate showing the contractor's model number, NSN, serial number, USA registration number, date of manufacture, and date of delivery. The identification plate shall be permanently affixed to the loader in a readily accessible and conspicuous location. Plates shall conform to MIL-P-514 or

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contractor's standard plates subject to the approval of the contracting officer. All attachments or components removed or disassembled for shipment shall be match marked for proper reassembly. The loader and each attachment shall be identified conforming to the contractor's standard method.

3.34.1 Instruction plates. The loader shall be equipped with instruction plates, including warnings and caution suitably located and permanently affixed, describing any special or important procedure to be followed in operating and servicing the loader. Plates shall conform to MIL-P-514 or contractor's standard plates subject to the approval of the contracting officer.

3.34.2 Shipping data plates. Shipping data plates for each loader shall be furnished and shall show the silhouette of the loader indicating the location and capacity of slinging provisions, tiedown provisions, the center of balance and shipping weight. Plates shall conform to MIL-P-514 or contractor's standard plates, subject to the approval of the contracting officer. The plates shall be attached to the left side of the loader and shall show the location when viewed from the left side.

3.34.3 Vehicle weight classification sign kit. The contractor shall apply the vehicle weight classification number to each vehicle utilizing a vehicle weight classification sign set (see 3.37). The set shall be bolted on the front of the loader in a location approved by the contracting officer. The contracting officer shall assign the classification numbers to be displayed on the loader.

3.34.4 Stenciling. Stenciling instructions shall be provided with each set of technical publications. The instructions shall be provided in card form showing location and wording for tiedown and slinging provisions, tire pressure, and the center of gravity. Each loader shall be stenciled, using the instructions provided with the stencil publications.

3.35 Government-furnished property. The following property in the quantity indicated will be furnished by the Government (see 6.4).

Item No.		Description Quantity
1.	Camouflage paint pattern.	As required.
2.	Case, NSN 7520-00-559-9618, complete with maintenance forms. (The contractor shall furnish the fasteners and install the case on the loaders.)	As required.
3.	Vehicle weight classification sign set, NSN 9905-00-565-6267.	As required.
4.	Report CR 82-588-003 STE/ICE Design Guide for Vehicle Diagnostic Connector.	As required.

3.36 Basic issue item. When specified (see 6.2), each item in the basic issue item list shall be furnished with each loader.

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3.37 Vehicle classification. When specified (see 6.2), the contractor shall complete and submit vehicle classification (figure 1).

3.38 Workmanship. Defective components shall not be furnished. Parts and assemblies which have been repaired or modified to overcome deficiencies shall not be used. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Boltholes and rivet holes shall be accurately punched or drilled and shall have burrs removed. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member. The surface of parts to be welded shall be free from rust, scale, paint, grease or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the weld.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection. The first article loader furnished by the contractor shall be examined (see 4.3.1), and tested (see 4.3.2) to determine compliance with the requirements of this specification. Prior to the examination and testing of the loader, the following shall be performed:

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a. Service the loader with the lubricants specified.

b. Break in the loader as prescribed by the contractor.

4.3.1 First article examination. Prior to the first article test, the loader shall be examined for the defects marked "X" in column 1 of table II. Presence of one or more defects shall be cause for rejection. Prior to the first article test, the loader shall be examined and compared with this specification, with the contractor's published brochures, and with the complete Technical Information Package (see 6.7). Any discrepancy between the specification, brochures, or Technical Information Package and the first article model shall be cause for rejection. Prior to the first article test, the calculations, certifications, and certified test reports required by 4.5.2.21 4.5.2.22 and 4.5.2.23 shall be examined. Incomplete, inaccurate, or otherwise unacceptable calculations, certifications, or certified test reports shall be cause for rejection.

4.3.2 First article tests. Upon successful completion of the examination specified in 4.5.1, the first article loader shall be subjected to tests marked "X" in column 1 of table III. Failure of any test shall be cause for rejection. All first article tests shall be run using MIL-T-83133, grade JP-8 fuel.

4.4 Quality conformance inspection.

4.4.1 Individual examination. After successful completion of the test specified in 4.4.2, each loader shall be examined for the defects marked "X" in column 2 of table II. Presence of one or more defects shall be cause for rejection.

4.4.2 Individual tests. Each loader shall be subject to the tests marked X in column 2 of table III. Failure of any test shall be cause for rejection. The production run in test (see 4.5.2.8) shall be run on either VV-F-800, grade DF-2 or MIL-T-83133, grade JP-8 fuel.

4.5 Inspection procedure.

4.5.1 Examination. The examination shall conform to table II.

TABLE II. Examination schedule.

First article 1	Individual 2	Defect 3	Requirement paragraph 4
X	X	101. Loader not as specified.	3.1
X	-	102. First article loader not as specified.	3.2
X	X	103. Assembly incomplete or incorrect. Material not of same quality as first article model.	3.3
X	X	104. Materials not as specified.	3.4

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TABLE II. Examination schedule. (Cont'd)

First article 1	Individual 2	Defect 3	Requirement paragraph 4
X	X	105. Material deterioration prevention and control not as specified.	3.4.1
X	X	106. Dissimilar metals not as specified.	3.4.1.1
X	X	107. Identification of materials not as specified.	3.4.1.2
X	X	108. Used, rebuilt or remanufactured components, pieces or parts incorporated in the loader.	3.4.2
X	X	109. Cab not as specified.	3.5
X	X	110. Controls not as specified.	3.5.1
X	X	111. ROPS and FOPS mounting not as specified. Seatbelt not as specified.	3.6
X	X	112. Fenders not as specified.	3.7
X	X	113. Seat not as specified.	3.8
X	X	114. Horn not as specified.	3.9
X	X	115. Backup alarm not as specified.	3.10
X	X	116. Diesel engine not as specified. Exhaust system not as specified.	3.11
X	X	117. Engine horsepower not as specified.	3.11.1
X	X	118. Engine side panels not as specified.	3.11.2
X	X	119. Air cleaner not as specified.	3.11.3
X	X	120. Engine cooling system not as specified.	3.11.4
X	X	121. Tachometer drive not as specified.	3.11.5
X	X	122. Oil sampling valve not as specified.	3.11.6
X	X	123. Drive train not as specified.	3.12
X	X	124. Transmission not as specified.	3.12.1

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TABLE II. Examination schedule - Continued.

First article 1	Individual 2	Defect 3	Requirement paragraph 4
X	X	125. Torque converter not as specified.	3.12.2
X	X	126. Drive axles not as specified.	3.12.3
X	X	127. Fuel capacity and fuel tank not as specified.	3.13
X	X	128. Steering system not as specified.	3.14.1
X	X	129. Braking system not as specified.	3.14.2
X	X	130. Air brake system not as specified.	3.14.2.1
X	X	131. Hydraulic system not as specified. Boom kickout not as specified.	3.14.3
X	X	132. Tires not as specified.	3.15
X	X	133. Electrical system not as specified.	3.16
X	X	134. Lighting not as specified.	3.16.1
X	X	135. Alternator not as specified.	3.16.2
X	X	136. Instruments and indicators not as specified.	3.16.3
X	X	137. Battery and battery enclosure not as specified.	3.17, 3.17.1
X	X	138. Slaving components not as specified.	3.17.2
X	X	139. Fungus- and moisture-resistance not as specified.	3.17.3
X	X	140. Towing device not as specified.	3.18
X	X	141. Multipurpose bucket not as specified.	3.19.1
X	X	142. Counterweight and tire ballast not as specified.	3.19.2
X	X	143. Special tools not as specified.	3.20
X	X	144. Toolbox not as specified.	3.20.1
X	X	145. Blackout lighting not as specified.	3.21.1

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TABLE II. Examination schedule - Continued.

First article 1	Individual 2	Defect 3	Requirement paragraph 4
X	X	146. Winterized loader not as specified.	3.21.2
X	X	147. Winterization kit not as specified, when applicable.	3.21.3
X	X	148. Kit installation not as specified.	3.21.4
X	X	149. Insulation not as specified, when applicable.	3.21.5
X	X	150. Heater and heat exchanger not as specified, when applicable.	3.21.6
X	X	151. DCA not as specified.	3.21.8
X	X	152. Antivandalism protection not as specified.	3.22
X	X	153. Lubrication not as specified.	3.23, 3.23.1
X	X	154. Human factors not as specified.	3.25
X	X	155. Safety items not as specified.	3.26
X	X	156. Noise limits not as specified.	3.27
X	X	157. Loader not furnished with equipment to enable loader to start and operate in ambient temperatures from -25 ^{0}F to +120 ^{0}F.	3.28
X	X	158. Loader not furnished with equipment to enable loader to be stored for extended periods at temperatures from -30 ^{0}F to +160 ^{0}F. Cold start aid not as specified.	3.28
X	X	159. Decontamination unit not as specified.	3.29
X	X	160. Electromagnetic radiation limits not as specified.	3.30
X	-	161. Transportability not as specified.	3.31
X	-	162. Tiedown provisions not as specified.	3.31.1

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TABLE II. Examination schedule - Continued.

First article 1	Individual 2	Defect 3	Requirement paragraph 4
X	-	163. Slinging provisions not as specified.	3.31.2
X	-	164. Air transportability requirements not met.	3.31.3
X	-	165. Rail transportability requirement not met.	3.31.4
X	-	166. Highway transportability requirements not met.	3.31.5
X	X	167. Treatment and painting not as specified. Camouflage pattern data not as specified. Color not as specified.	3.32
X	X	168. Camouflage pattern data not provided when specified.	3.32.1
X	X	169. Dimensions not as specified.	3.33.1
X	X	170. Weight not as specified.	3.33.2
X	X	171. Identification plates not as specified.	3.34
X	X	172. Instruction plates not as specified.	3.34.1
X	X	173. Shipping data plates not as specified.	3.34.2
X	X	174. Vehicle weight classification sign kit not as specified.	3.34.3
X	X	175. Stenciling instructions not as specified.	3.34.4
X	X	176. Government-furnished property not installed on loader.	3.35
X	X	177. Basic issue items not furnished when specified.	3.36
X	-	178. Vehicle classification.	3.37

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TABLE II. Examination schedule - Continued.

First article 1	Individual 2	Defect 3	Requirement paragraph 4
X	X	179. Parts or components missing. Assembly incomplete or incorrect, with cracked or broken parts. Permanent deformation of any part or assembly.	3.38

4.5.2 Tests. Tests shall be conducted conforming to table III.

4.5.2.1 Test Schedule.

TABLE III. Test schedule.

First article model	Individual	Test	Test paragraph	Requirement paragraph
X	-	Construction task.	4.5.2.2	3.1, 3.13, 3.24
X	-	ROPS and FOPS.	4.5.2.3	3.6
X	-	Braking system.	4.5.2.4	3.14.2
X	-	Installation test, when applicable.	4.5.2.5	3.21.4
X	-	Cold test, when applicable.	4.5.2.6	3.21.7
X	-	DCA capability.	4.5.2.7	3.21.8
X	-	DCA-STE/ICE interfaces.	4.5.2.7.1	3.21.8
X	X	Production run-in.	4.5.2.8	3.24
X	-	Weight.	4.5.2.9	3.33.2
X	-	Operating and tipping load.	4.5.2.10	3.24.1, 3.24.2
X	-	Lift capacity.	4.5.2.11	3.24.3
X	-	Breakout force.	4.5.2.12	3.24.4
X	-	Clam closure force.	4.5.2.13	3.24.5
X	-	Drawbar pull.	4.5.2.14	3.24.6
X	-	Travel speed.	4.5.2.15	3.24.7
X	-	Cycle time.	4.5.2.16	3.24.8
X	-	Clearance circle.	4.5.2.17	3.24.9
X	-	Stability.	4.5.2.18	3.24.10
X	-	Fording.	4.5.2.19	3.24.11
X	-	Noise level	4.5.2.20	3.27

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TABLE III. Test schedule - Continued.

First article model	Individual	Test	Test paragraph	Requirement paragraph
		test.		
X	-	Tiedown provisions.	4.5.2.24.1	3.31.1
X	-	Slinging provisions.	4.5.2.24.2	3.31.2
X	-	Rail impact test.	4.5.2.24.3	3.31.4

4.5.2.2 Construction tasks. The loader shall be tested as follows for a total of 100 hours to prove conformance to 3.1 and 3.13.

a. Using bank run gravel or building sand, test for 35 hours loading (from stockpiles) dump trucks having a sideboard height of not less than 80 inches above ground level. Approximately one-half of this operation shall be conducted with the load bottom dumped.

b. Using bank run gravel or building sand, test for 35 hours the loader's ability to move stockpiles with capacity loads a distance of not less than 100 feet. Approximately one-half of this operation shall be conducted with load bottom dumped.

c. Test for 15 hours the loader's ability to excavate a pit in undisturbed soil using the bucket closed.

d. Test for 15 hours the loader's ability to excavate a pit in undisturbed soil using the open bucket as a dozer.

Nonconformance to 3.1 and 3.13 or any malfunction or breakage of mechanical or electrical components shall constitute failure of this test.

4.5.2.3 ROPS and FOPS removal and replacement. The removal and replacement of the ROPS and FOPS structure shall be tested following procedures in the contractor's manual. Permanent deformation or damage to any part of the ROPS and FOPS structure or its mounts, or inability to meet the requirements of 3.6 and 3.31 shall constitute failure of this test.

4.5.2.4 Braking systems. The braking systems of the loader shall be tested to prove conformance to 3.14.2. Nonconformance to 3.14.2 shall constitute failure of this test.

4.5.2.5 Installation test. The purpose of this test is to ensure the proper fit, form and function of the winterization kit so that no operational or safety deficiencies exist after kit installation on the loader. The loader shall be demonstrated for 10 hours performing 50 percent of the operations identified in 4.5.2.2 to prove conformance to 3.21.4.

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4.5.2.6 Cold tests. The loader shall be cold soaked in a temperature-controlled chamber to a temperature of -50 ^{0}F and tested as follows to prove conformance to 3.21.7.

a. Preparation.

(1) Ensure that the loader has been serviced with arctic coolant, fuels, lubricants, and hydraulic oils.

(2) Service Swingfire heater with arctic gasoline.

(3) Place loader in temperature-controlled chamber.

(4) Record, as a minimum, the following temperatures:

(a) Exhaust: engine heater outlet.

(b) Air: battery boxes, inlet, outlet, and all sides.
(c) Air: operator's compartment and operator torso (seated position).
(d) Air: operator's compartment, floor, roof, and sides (midway to top).

(e) Air: hot-water-heater outlet.

(f) Electrolyte: all batteries.

(g) Coolant: engine block inlet.

(h) Coolant: engine block outlet.

(i) Oil: engine sump and gallery.

b. Procedure.

(1) Subject loader to a stabilized temperature of -50 ^{0}F for 24 hours.

(2) Bring Swingfire heater into cold chamber. Connect heater to loader electrical power source and start heater. (After heater start, disconnect from power source.)

(3) Insert heater into heat exchanger and preheat engine coolant and batteries for 90 minutes.

(4) Record temperatures continuously or at intervals not greater than 5 minutes.

(5) Remove heater from heat exchanger and shutoff motor.

(6) Start engine from loader batteries.

(7) Start personnel heater and defrost fans 30 minutes after engine start.

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(8) One (1) hour after heater start, record operator compartment air temperatures and check defrost.

(9) Operate loader steering and bucket controls for 10 minutes within the dimensional limits of the test chamber.

(10) Engage and operate transmission and brake system within the dimensional limits of the test chamber.

Nonconformance to 3.21.7 shall constitute failure of this test.

4.5.2.7 DCA capability. The contractor will perform a 100- percent visual inspection of all DCA wiring harnesses prior to installation on the loader to ensure wires and connections are free of error. After installation on the loader, test the DCA harness (reference chapter 5 of design guide for DCA) on each vehicle with a DCA tester prior to interfacing with STE/ICE. Nonconformance to 3.21.8 shall constitute failure of the DCA.

4.5.2.7.1 DCA - STE/ICE interfaces. Test the DCA for all measurements specified in 3.21.8 using STE/ICE. After completion of all first article testing, the test shall be repeated and the results compared to the first test performed as specified in 4.5.2. Nonconformance to 3.21.8 or failure of any component or wiring in the DCA harness, which results in inability to obtain the measurements previously recorded with the STE/ICE (see 4.5.2), shall constitute failure of this test.

4.5.2.8 Production run-in. Start and operate each loader for not less than 0.50 hour to assure operation of all components and to allow for minor adjustments. Start and stop the engine not less than five times. Operate all bucket controls for 5 minutes. Hold each hydraulic control in the actuation position until the hydraulic system reaches relief valve pressure at least five times. Operate the loader at least two times through a figure eight steering course at the maximum safe speed. Make not less than five abrupt panic stops from a speed not less than 19 mph using the service brakes. Release and reapply the parking brakes not less than five times. Turn on all lights for the duration of the run-in. Evidence of any deficiency, need for adjustment or maintenance, malfunction of any control, or evidence of leakage of fuel, lubricants, or hydraulic fluid shall be cause for rejection of the loader by the Government until corrected to the Government's satisfaction. The Government may require additional run-in after deficiencies are corrected. Any defect found during production run-in shall be prima facie evidence that loaders previously accepted are similarly defective unless evidence, satisfactory to the contracting officer, is furnished by the contractor that they are not similarly defective.

4.5.2.9 Weight. The weight of the loader shall be measured with a scale calibrated within 6 months of the tests. Inability to meet the requirements of 3.31.3 and 3.33.5 shall constitute failure of this test.

4.5.2.10 Operating and tipping load. The operating and tipping load of the loader shall be measured to prove conformance to 3.24.1 and 3.24.2. Nonconformance to 3.24.1 and 3.24.2 shall constitute failure of this test.

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4.5.2.11 Lift capacity. The lift capacity at the carry position and the lift capacity to maximum height shall be measured to prove conformance to 3.24.3. Nonconformance to 3.24.3 shall constitute failure of this test.

4.5.2.12 Breakout force. The breakout force of the loader shall be measured to prove conformance to 3.24.4. Nonconformance to 3.24.4 shall constitute failure of this test.

4.5.2.13 Clam closure force. The clam closure force of the loader shall be measured to prove conformance to 3.24.5. Nonconformance to 3.24.5 shall constitute failure of this test.

4.5.2.14 Drawbar pull. The drawbar pull of the loader shall be measured to prove conformance to 3.24.6. Nonconformance to 3.24.6 shall constitute failure of this test.

4.5.2.15 Travel speed. The travel speed of the loader shall be measured to prove conformance to 3.24.7. Nonconformance to 3.24.7 shall constitute failure of this test.

4.5.2.16 Cycle time. The raising time, lowering time, and dump time of the loader shall be measured to prove conformance to 3.24.8. Nonconformance to 3.24.8 shall constitute failure of this test.

4.5.2.17 Clearance circle. The clearance circle of the loader shall be measured to prove conformance to 3.24.9. Nonconformance to 3.24.9 shall constitute failure of this test.

4.5.2.18 Stability. The stability of the loader shall be measured to prove conformance to 3.24.10. Nonconformance to 3.24.10 shall constitute failure of this test.

4.5.2.19 Fording. The fording ability of the loader shall be demonstrated to prove conformance of 3.24.11. Nonconformance to 3.24.11 shall constitute failure of this test.

4.5.2.20 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load. The noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.21 Calculation. The contractor shall furnish calculations to prove conformance to the bucket capacity requirements of 3.19.1 and the air transportability requirements of 3.31.3. Dimensions used in the calculations shall be verified by measurement. Nonconformance to 3.19.1 or 3.31.3 shall be cause for rejection.

4.5.2.22 Certification. The contractor shall certify in writing to the contracting officer that the loader complies with the starting and operating temperature requirements of 3.28, the engine requirements of 3.11, the horsepower requirement of 3.11.1, the steering requirement of 3.14.1, and the

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rail shipment requirements of 3.31 and 3.31.4. If certification is not supplied, these requirements shall be tested. Nonconformance shall constitute failure of this test.

4.5.2.23 Certified test reports. The contractor shall furnish certified test reports to prove compliance with the ROPS and FOPS requirements of 3.6 and electromagnetic radiation of 3.30. If certified test reports are not available, these requirements shall be tested. Nonconformance shall constitute failure of these tests.

4.5.2.24 Transportability.

4.5.2.24.1 Tiedown provisions. The tiedown provisions shall be tested conforming to MIL-STD-209 to prove conformance to 3.31.1. Inability to met the requirements of 3.31.1 shall constitute failure of this demonstration.

4.5.2.24.2 Slinging provisions. The slinging provisions shall be tested conforming to MIL-STD-209 to prove conformance to 3.31.2. Inability to meet the requirements of 3.31.2 shall constitute failure of this demonstration.

4.5.2.24.3 Rail impact test. The first article loader shall be tested conforming to the following rail impact test procedure:

Step 1. From one to five railcars shall be positioned on a level section of track. The following conditions shall be satisfied:

a. The total weight of the cars shall be at least 250,000 pounds (114,000 kilograms).
b. The couplers between any cars shall be compressed to take up any slack.
c. All the air and hand brakes on the cars used shall be set.
d. Any load in or on the cars shall be secured to prevent sliding or shifting; any movement greater than 1.97 inches (5 centimeters) shall be justification for retest.
e. The end of the buffer car to be struck must have a standard draft gear.

Step 2. Mount the test item on the test car. The test car shall be equipped with standard draft gears and conventional under frames. Mounting of the test item shall incorporate the standard loading and bracing method as shown in section 6 of the Association of American Railroads (AAR) Rules Governing the Loading of Department of Defense Materiel on the open Top Cars. No exotic or unusual tiedown methods shall be used; any non-standard loading and bracing must be approved by the MTMC/TEA prior to testing. The arrangement of the test item and its blocking and tiedown to be tested shall be identical to that proposed and approved by MTMC/TEA (if non-standard).

Step 3. Situate the test car between the buffer cars and the locomotive, and pull the test car at least 200 feet (65 meters) from the buffer cars along a level section of track (a minimum distance to achieve the required locomotive speeds).

Step 4. Position the knuckles of the buffer and test cars for coupling.

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Step 5. Install a timing device to measure the test car speed (- or - .1 km/hr) just prior to impact with the buffer cars. Suggested methods include electronic timing (microswitches) and radar. The use of torpedoes or a stopwatch is permissible but not recommended because of the inaccuracies involved.

Step 6. Push the test car toward the buffer cars and, by using the locomotive's speedometer or other means, release the test car when the desired test speed is reached, thus allowing the test car to freely impact the buffer cars.

Step 7. Repeat step 6 until the test cars has impacted the buffer car three times at the same end, once each at speed of 4, 6, and 8, +0.5, -0.0 mph (5.4, 9.7 and 13, +0.8, -0.0) km/hr. Reverse the test car and repeat the 8 mph (13.0 km/hr) for a total of 4 impacts.

NOTE: Adjustment of the loading or securing mechanism or reconditioning of the bracing or items of securement are not allowed without a complete retest.

4.5.2.24.4 Failure criteria. Failure to meet the requirements of 3.31.4 and 4.5.2.24.3 rail impact test shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Examination. Each loader, prepared for shipment, shall be examined for the following defects.

180.	Preservation not as specified for level A, B or C (see 5.1).
181.	Packing not as specified for level A, B or C (see 5.2).
182.	Marking incorrect, illegible or missing (see 5.3).

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each complete loader, components, and attachments shall be preserved conforming to level A preservation and packaging requirements of MIL-T-3351.

5.1.2 Level B. Each complete loader, components, and attachments shall be preserved conforming to level B preservation and packaging requirements of MIL-T-3351.

5.1.3 Level C. Each complete loader and all items issued with the loader shall be preserved conforming to MIL-P-116, method III. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservation coating, volatile corrosion inhibitors, or desiccated packs. Items requiring protection from physical damage to surfaces, or which are fragile by nature shall be protected by wrapping, pack compartmentization, or cartonizing of

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the individual item. The cooling system shall be filled with a fresh solution of 50 percent water and 50 percent antifreeze, conforming to MIL-A-46153 and a tag attached indicating the temperature to which the cooling system can be subjected before freezing. The crankcase shall contain the required amount of lubricating oil. A guide shall be furnished indicating the preservation and packing materials to be removed, in detail, and any other servicing required prior to placing the loader in operation.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Each complete loader, preserved as specified in 5.1, shall be packed conforming to level A packing requirements of MIL-T-3351, as specified for type II, unboxed (mobile) shipment.

5.2.2 Level B. Each complete loader, preserved as specified in 5.1, shall be packed conforming to level B packing requirements of MIL-T-3351, as specified for type II, unboxed (mobile) shipment.

5.2.3 Level C. Each complete loader, preserved as specified in 5.1, shall be packed for mobile shipment to protect against physical and environmental damage during shipment, handling, and storage. Boxed accessory items shall be secured in a protected location to the loader.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall conform to MIL-STD-129.

6. NOTES.

(This section contains information of a general or explanatory nature which is helpful, but is not mandatory.)

6.1 Intended use. The loader is intended to be used primarily for excavating, digging, loading, stripping, leveling, stockpiling, and transferring of earth and aggregate.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Timeframe for submission of first article model (see 3.2).

d. When nonstandard tools are not required (see 3.20).

e. When winterized loaders are required and number of loaders to be furnished (see 3.21.2).

f. Timeframe required for submission of winterized loader for examination and test (see 3.21.2).

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g. When Swingfire heaters or their equal are required and number of heaters to be furnished (see 3.21.6).

h. When DCA is not required (see 3.21.8).

i. When air transportability data is requested for the Technical Information Package (see 3.31.3).

j. Color required when other than as specified (see 3.32).

k. When overcoating with camouflage pattern is required (see 3.32).

l. When camouflage patterns will be provided (see 3.32.1 and 6.4).

m. When basic issue items are to be furnished (see 3.36).

n. Timeframe for completing and submitting the vehicle classification (see 3.37).

o. Level of preservation and packing required (see 5.1 and 5.2).

p. Any special marking (see 5.3).

q. When a TIP is required and timeframe for submission of TIP (see 6.7).

r. When transportability report required (see 6.8).

6.2.2 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

	Ref. Para.	DID Title	DID No.	Suggested Tailoring
a.	3.32.1	Color photograph prints		DI-MISC-80192
b.	3.32.1	Camouflage line art data		DI-MISC-80176
c.	3.21.8	Commercial-off-the-shelf (COTS) Manual	DI-TMSS-80527	

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

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(Copies of DIDs required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.3 First article. When a first article inspection is required, the item should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the first article. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.3.5.

6.5 Lubricants. MIL-STD-838, Lubrication of Military Equipment, prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of nonstandard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113, Guide for the Selection of Lubricants, Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development and Engineering Center (ATTN: STRBE-VF), Fort Belvoir, VA 22060-5606, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.6 Vehicle classification. The contracting officer should arrange to furnish a vehicle classification specification sheet to the contractor and request the return of reviewed sheet to the Government. The information thus obtained should be utilized to determine the weight classification numbers for the loader. The contracting officer should then request the contractor to apply the numbers to the loader. The use of a Government-furnished sign kit conforming to MIL-S-40626 should be utilized where applicable (see 3.34.3).

6.7 Technical information package (TIP). When required (see 6.2), the contractor shall complete and submit a TIP within the timeframe specified depicting the loader and its components exactly as it will be furnished. Notation shall be made of all components which have not been in field use for a minimum of 1 year and shall include identification of the component or part

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previously used. The TIP shall be submitted with all bids and proposals. Copies of the TIP are available from the Belvoir Research, Development and Engineering Center, ATTN: STRBE-FMT, Fort Belvoir, VA 22060-5606.

6.8 Transportability report. When specified (see 6.2), the contracting officer should arrange for the contractor to furnish a transportability report conforming to AR 70-47 and S.O.P. 70-38 for the loader 95 days prior to delivery of the first article loader.

6.9 AQL certification. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQL's/LTPD's (Acceptable Quality Levels/ Lot Tolerances Percent Defectives) from military specifications.

(Certified by/date: 9/20/89).

6.10 Subject term (key word) listing.

Loader, articulated frame
Loader, diesel engine
Loader, four-wheel-drive
Loader, scoop type

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activities:
DLA - CS

Project 3805-A123

User activity:
Army - AT, CE

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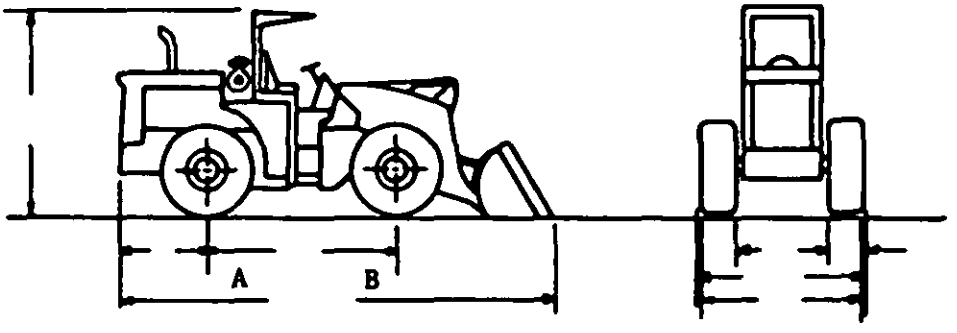
Load Class

E	C	H

NOMENCLATURE:

Mfr.		Model No.
NSN	Agency	Stock No.

Traveling Load Category		Total Weight W	Pay Load
Empty	E		X
Loaded	C		



Traveling Load Category	LOADS ON		Front Tires			Rear Tires		
	A	B	No/ axle	size	psi	No/ axle	size	psi
Empty								
Loaded								

All loads in short tons.
All loads to include weight of crew and appurtenances.
Class No's. for vehicles over 3,000 lbs.
Longitudinal dimensions in feet.
Vertical and transverse dimensions in inches.

FIGURE 1. Vehicle classification data sheet.

X-3772A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-52385D(ME)	Loader, Scoop Type, DED, 4x4, Articulated Frame Steer, 2-1/2 Cubic Yard, Commercial Construction Equipment

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 32 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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